KANCHANI RAJYALAKSHMI

Software Engineer
kanchanirajyalakshmi@gmail.com | +91 8143688131 | Andhra Pradesh, India
linkedin.com/in/rajyalakshmi-kanchani-39781b34a

PROFESSIONAL SUMMARY

B.Tech Computer Science student (Class of 2027) with software engineering internship experience building backend and product features for a cloud-native startup. Hands-on with Python, FastAPI, React, SQL and PostgreSQL, and building AI applications with LLM APIs, retrieval-augmented generation (RAG), embeddings and tool calling. Comfortable working independently in a remote setting and collaborating through pull requests, code reviews and agile sprints. ServiceNow certified and eager to keep learning technologies that solve real-world problems.

CORE SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS

Frameworks & Tools: FastAPI, React, REST APIs, Git, GitHub, Postman, VS Code

AI / ML: LLM APIs, Prompt Engineering, RAG, Embeddings, Vector Search, Tool Calling, Evaluation

Databases: PostgreSQL, SQL

Core CS: OOPs, DBMS, Operating Systems, Data Structures & Algorithms

Practices: Code Reviews, Documentation, Debugging, Software Testing, Agile Workflows, Remote Collaboration

PROFESSIONAL EXPERIENCE

Software Engineering Intern – UZO (Startup) *Apr 2026 – Jun 2026*

- Worked with the engineering team at UZO, a cloud-native platform startup, contributing to backend and product features.
- Built and tested REST API endpoints using Python and FastAPI, validating requests and responses with Postman.
- Collaborated through Git and GitHub using pull requests and code reviews, and followed agile sprint workflows.

PROJECTS

AI Healthcare Assistant

- Built a RAG-based healthcare assistant that answers questions from approved medical information with source citations.
- Implemented authentication and role-based access controls using FastAPI and PostgreSQL to protect user data.
- Added safe escalation that detects medical-emergency queries and directs users to emergency services instead of answering.
- Evaluated answer quality and deployed the application.

Research Assistant

- Created an agent workflow that searches approved sources, summarizes evidence and produces structured reports with cited references.
- Used LLM tool calling to orchestrate search, extraction and summarization steps, with evaluation checks on report accuracy.

PDF Question-Answering Assistant

- Built an assistant that lets users upload PDFs and ask questions, returning answers with page-level source references.
- Implemented text extraction, chunking, embedding generation and vector search to retrieve the most relevant passages.

GeneWeaver: GPU-Accelerated CRISPR Alignment Engine

- Developed a GPU-accelerated CRISPR off-target alignment engine to process large-scale genomic datasets for DNA sequence analysis.
- Engineered CUDA JIT kernels and optimized GPU memory management to accelerate sequence alignment through parallel execution.
- Designed a Dask pipeline for multi-GPU scheduling and a Textual dashboard to monitor GPU usage and alignment progress.

EDUCATION

B.Tech, Computer Science and Engineering *2027 | Kavali, Andhra Pradesh*

PBR Visvodaya Engineering College *CGPA: 86*

CERTIFICATIONS

- ServiceNow Certified System Administrator
- ServiceNow Certified Application Developer
- freeCodeCamp Scientific Computing with Python